

Generative AI and Labor Market Outcomes: Evidence from the United Kingdom

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Abstract

This paper examines the effects of LLMs on UK labor market outcomes. We use a difference-in-differences design that compares outcomes across firms and occupations from 2021 to 2025 based on their differential exposure to LLM capabilities. Our results show that highly exposed firms reduce employment, with effects concentrated in junior positions. These firms sharply curtail new hiring, with technical and creative roles experiencing the steepest declines. At the occupation level, roles more exposed to LLMs also show substantial reductions in job listings. The observed effects concentrate almost entirely in high-wage segments. Hence, our findings suggest that LLMs may compress wage inequality.

Keywords: Generative AI; Large Language Models; Automation; Labor demand; United Kingdom.

JEL: O33; J23; J31

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sample construction by varying minimum firm size thresholds. We consider alternative task-level weights to construct our exposure measures. We estimate weighted regressions that give greater weight to larger firms and occupations to ensure our results reflect typical worker experiences. Finally, we relax the assumption of linear treatment effects by categorizing firms and occupations into exposure quartiles. These analyses consistently support our main findings.

These robustness checks notwithstanding, we acknowledge the possibility that other factors differentially affected exposed and less exposed firms or occupations around the time of ChatGPT’s introduction. For instance, if firms anticipating AI disruption had already begun adjusting their workforce composition before November 2022, or if other technological or economic shocks coincidentally affected high-exposure entities differently, our estimates could be biased. Additionally, our exposure measures capture potential susceptibility to LLM disruption but may not perfectly predict actual adoption patterns, which likely vary with firm-specific factors such as technological capabilities, management quality, and financial resources.

4 Results

This section presents our main empirical findings on the effects of LLM exposure on UK labor market outcomes. We organize the analysis in three parts. First, we examine how employment stocks, workforce composition, and compensation evolve within firms with different levels of LLM exposure. Second, we investigate occupation-level hiring patterns through job posting data. Third, we explore within-firm recruitment strategies by combining firm-level exposure measures with their occupation-specific hiring patterns. Throughout, we present both event study specifications and static difference-in-differences estimates.

4.1 Firm-Level Employment Effects

Figure 1 presents the event study plots for the effect of LLM exposure on the total number of employees in UK firms. We find that highly exposed firms start experiencing a statistically significant decline in total employment beginning approximately 18 months after the introduction of ChatGPT, suggesting that exposed firms reduce headcount in response to this new technology, but that it takes time to